

Multi-Factor ETF Strategy

Operating Document

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Live account: U9544585

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Abstract

Long-only smart-beta ETF strategy operated against a live IB Gateway account. Selects 30 positions from a 599-ticker hand-curated universe (leveraged / inverse / commodity / currency / volatility excluded) by weighted geometric mean of four AQR-canonical factors: momentum 35%, quality 30%, low-volatility 20%, value 15%. Exponential rank weighting, 2%–15% position bounds, 10% trailing stops, bimonthly rebalance with 5% drift trigger. **Multi-regime validated over 15.5 years (2011–2026): CAGR +13.0%, Sharpe +0.51, MaxDD −37%; beats SPY same-period CAGR by ~0.9%/yr at comparable drawdown.** This document is the operator's reading surface; the code in `src/` is authoritative for behaviour, this document for intent.

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1 Executive Summary

1.1 What This Is

A systematic, deterministic, long-only ETF portfolio. No LLM sets sizes or places orders. No factor timing. No sentiment trading. Every numeric parameter traces to either academic literature or a repo-recorded backtest.

Operated weekly: cache refresh Saturday evening, review Sunday, apply mid-week when the model proposes a rebalance and the operator approves.

1.2 Performance — headline KPIs

Metric	Full 15.5y	Trailing 5y	Trailing 3y
CAGR	+13.04%	+12.44%	+19.31%
Time-average log-CAGR	+12.28%	—	—
Sharpe ratio	+0.51	+0.52	+0.97
Sortino ratio	+0.62	+0.69	+1.30
Max drawdown	−36.9%	−19.1%	−13.4%
Annualised volatility	20.3%	—	—
Avg turnover per rebalance	~50%	—	—
Alpha vs SPY (CAGR)	+0.89%	—	—
SPY same-period reference	+12.15% CAGR / −34.1% MaxDD	—	—

Backtest window 2011-01-28 to 2026-04-30 (3 904 trading days), 599-ticker curated universe, 30-position basket, exponential rank weights, 5 bps turnover cost. Full methodology and per-window detail in §4 and docs/research/2026-07_multi_regime_results.md. Multi-regime coverage includes EU crisis 2011, taper tantrum 2013, China 2015–16, 2018 Q4, COVID 2020, and 2022 rate cycle.

1.3 Validation — one line each

- **Factor weights 35/30/20/15:** empirically validated. Best grid variant beats prior by +0.024 Sharpe — well inside the pre-registered ≥ 0.10 change threshold. Prior sits inside a broad plateau of near-tied configurations. *Retained.*
- **Real value factor:** rewired 2026-07-10 to real per-fund yield + expense ratio from FMP Premium. No longer a fee proxy. Rankings verified (SCHD > HDV > VYM > VTV > ... > VUG > VGT > ARKK).
- **Universe:** 599-ticker hand-curated smart-beta list. Leveraged / inverse / commodity / currency / volatility all excluded, applied identically in the live pipeline and the diagnostic.
- **Regime overlay (SPY 200d + VIX):** tested, not auto-enabled. Improves early-crisis drawdowns (2011, 2015) but underperforms baseline on the trailing 5-year window. Available as manual defensive toggle.
- **Correlation clustering:** tested, rejected. Every configuration made both CAGR and drawdown worse.

1.4 Live Snapshot (2026-07-09)

Net Liquidation Value	\$390,940
Cash	\$84,552
Positions	28
Live earnings (owner’s estimate)	~13–14%

Position count of 28 sits below the target of 30 (raised from 20 on 2026-07-10). The next rebalance will add positions and redeploy the freed \$40k from the reduced cash reserve.

1.5 Positioning

The strategy targets a modest, defensible CAGR premium over SPY at comparable drawdown — not a home run over the market. A year in which the strategy returns 12% while SPY returns 15% is a good year for the strategy *if* its downside is contained. The empirical validation above is what supports live deployment; the design principles in §3 are what keep it disciplined.

2 Strategy Mechanics

2.1 Universe Selection

The strategy runs on a hand-curated smart-beta universe declared in `src/data_collection/comprehensive_etf_universe` and enforced by `src.data_collection.etf_filters.filter_universe(use_curated=True)` in both the live pipeline and the multi-regime backtest diagnostic. Constructed once, revisited annually.

1. Hand-picked categories: US broad / mid / small, growth / value tilts, sector, dividend, international developed / emerging, bonds broad / credit, REIT, factor-multi.
2. Excluded categories: commodities, currency single-country, volatility products, thematic-single-narrative (cannabis, cannabis-only clean energy, etc.).
3. Excluded structure: leveraged / inverse / short (2x, 3x, ProShares Ultra, Direxion Daily) removed by enumerated list plus name-pattern matching.
4. History requirement: ≥ 252 trading days of price data before scoring at each rebalance date.
5. Result: **599 curated tickers** eligible for scoring at the 2026-07 snapshot. The screen is applied identically in the diagnostic and live pipeline so that backtest results simulate live behaviour.

2.2 Factor Framework — What the Four Factors Are and Why

The four factors below are the AQR-canonical smart-beta four: momentum, quality, low-volatility, value. They were chosen because they are the four with the largest and most robust empirical premia over the widest range of markets and time periods studied in the academic literature. AQR chose these because after decades of empirical work they are the return sources that survive out-of-sample testing, transaction costs, and factor-crowding. The strategy does not attempt to identify novel factors.

2.2.1 Momentum (target 35% weight)

$$\text{Momentum}_i = \frac{P_{i,t-21} - P_{i,t-252}}{P_{i,t-252}} \quad (1)$$

12-month return, with the most recent 21 days omitted. Winsorised at the 1st and 99th percentile.

Why: Assets that have outperformed over the prior year tend to continue outperforming over the next year. The 1-month skip avoids the short-term reversal effect that would contaminate the trend signal. Established by Jegadeesh & Titman (1993); the skip refinement from Novy-Marx (2012).

Why AQR uses it: The most-studied, most-robust factor after value, and the largest-premium factor in the last decade. Works across equity, futures, and FX. Fails during trend reversals (2022 tech drawdown, 2000 dot-com peak).

2.2.2 Quality (target 30% weight)

Composite of three z-score-normalised components on 252-day windows:

$$\text{Sharpe Ratio} = \frac{\mu_r - r_f}{\sigma_r} \sqrt{252} \quad (40\% \text{ weight}) \quad (2)$$

$$\text{Drawdown Resilience} = -1 \times \text{MaxDD} \quad (30\% \text{ weight}) \quad (3)$$

$$\text{Return Stability} = -1 \times \sigma_r \sqrt{252} \quad (30\% \text{ weight}) \quad (4)$$

Why: Assets that produce stable, high risk-adjusted returns tend to continue doing so. Established by Asness, Frazzini & Pedersen (2019).

Why AQR uses it: Quality captures the defensive-equity premium. Correlated with low-volatility but not identical — quality picks up profitability and stability of *returns*, low-vol picks up stability of *prices*. Both belong in the portfolio because they hedge different aspects of stress.

2.2.3 Low-Volatility (target 20% weight)

$$\text{Volatility}_i = \frac{1}{\sigma_{60d,i} \times \sqrt{252}} \quad (5)$$

Inverse of annualised realised volatility over 60 trading days. Higher score for lower-volatility ETFs.

Why: The low-volatility anomaly — lower-volatility assets have historically produced higher risk-adjusted returns than higher-volatility assets, contradicting the CAPM. Baker, Bradley & Wurgler (2011).

Why AQR uses it: Two reasons. First, the low-vol premium is real and persistent. Second, low-vol allocations reduce portfolio-level whipsaw, which matters more than arithmetic-mean metrics suggest in a compounding portfolio (see §3).

2.2.4 Value (target 15% weight)

$$\text{Value}_i = 0.60 \cdot z(\text{DividendYield}_i) + 0.40 \cdot z(-\text{ExpenseRatio}_i) \quad (6)$$

What each component contributes. Dividend yield is the canonical fund-level value tilt for ETFs — funds like SCHD, VYM, HDV, and VTV that hold value-tilted holdings publish materially higher trailing yields than growth-oriented funds like VUG, VGT, and QQQ. It is a real cross-sectional signal, not a fee preference. Expense ratio contributes the cost-efficiency component: for two funds with similar yield, the cheaper one is a better long-run compounder. Both components are z-score-normalised on the current cross-section before the blend.

Data source. Real per-fund dividend yield is computed as `lastDividend/price` from Financial Modeling Prep’s `stable/profile` endpoint. Real per-fund expense ratio comes from the `stable/etf/info` endpoint. Both are cached weekly to `~/trade_data/ETFTrader/processed/etf_fundamentals` by `scripts/refresh_fundamentals.py`. Rankings verified on 2026-07-10: SCHD 1.04, HDV

0.81, VYM 0.56, VTV 0.38, VOO -0.04, VUG -0.39, VGT -0.50, ARKK -1.81 — the expected value ordering.

Residual gap. Fund-level weighted P/E and P/B are still absent. The FMP endpoints that publish these (`key-metrics-ttm`, `ratios-ttm`) aggregate 10-K filings and return zero rows for ETFs at any subscription tier because ETFs do not file 10-Ks. Adding true weighted-average P/E would require holdings-level aggregation via `etf/holdings` (Ultimate tier) or a Morningstar-class source. Documented as an open enhancement in §4. Current implementation delivers approximately 60% of the intended value composite.

Why AQR uses value: It is the oldest and most-studied factor (Fama & French 1993). Tends to work when growth and momentum break (2000–2002, 2022). Yield-based value is a defensible, slowly-varying subset of the full academic tilt.

2.3 Factor Integration

$$\text{Score}_i = \text{Mom}_i^{0.35} \times \text{Qual}_i^{0.30} \times \text{Vol}_i^{0.20} \times \text{Val}_i^{0.15} \quad (7)$$

Each factor is first converted to a percentile rank in $[0,1]$. The integrated score is the weighted geometric mean of ranks.

Why geometric mean, not arithmetic: Geometric-mean integration penalises weakness on any one factor. An ETF in the 90th percentile on three factors but the 20th on one ranks below an ETF in the 70th percentile on all four. This “multi-factor consistency” requirement — explicitly the key insight from AQR’s factor-integration research — rejects one-trick candidates in favour of broadly-strong candidates. Arithmetic integration would allow one strong factor to compensate for weakness on the others.

2.4 Portfolio Construction

1. Score every ETF in the filtered universe.
2. Rank descending.
3. Take the top 30 (raised from 20 on 2026-07-10 — see Change Log).
4. Assign exponential rank weights: $w_i \propto e^{-\text{rank}_i/30}$, normalised to sum to 1.
5. Apply weight bounds: minimum 2%, maximum 15%.

Exponential weighting expresses conviction — the top-ranked position gets meaningfully more weight than the thirtieth — while the bounds prevent single-position concentration. The alternative optimiser (`OPTIMIZER_TYPE = "mvo"`) is available in the code but not the live default.

2.5 Rebalancing

1. Scheduled: bimonthly (~ 6 times per year).
2. Drift-triggered: if any position’s actual weight deviates from target by $>5\%$, rebalance.
3. In practice, drift triggers dominate; scheduled rebalances rarely fire independently.

Drift threshold is deliberately wide. Tighter thresholds cause more trading on smaller deviations — textbook whipsaw. See §2.8.

2.6 Risk Management

Every entry BUY places a 10% trailing stop, GTC, on the position. Trailing stops ratchet up as the position appreciates. On any 10% pullback from the running high, the stop triggers.

Entry stop-loss of 12% (a bit wider than the trailing stop) accommodates initial-position volatility before the trailing high-water mark establishes.

Cash reserve of \$30,000 is held unallocated (lowered from \$70,000 on 2026-07-10 to free capital for the widened 30-position basket). This is a static allocation, not a regime-responsive

hedge; the T2.1 regime overlay in progress will replace the static reserve with a dynamic risk-off multiplier.

2.7 Explicit Assumptions

The strategy assumes:

- Factor premia (momentum, quality, low-vol, value) will continue to exist over the holding horizon. They may attenuate but do not disappear.
- The rank structure of the universe changes slowly enough that monthly rebalancing captures the signal without significant slippage.
- Retail transaction costs at IB (~ 5 bps effective) are small enough not to erode the premium.
- Long-only holds; no short exposure to hedge market drawdown.
- No leverage.
- Trailing stops execute at or near their trigger prices. Gap-down scenarios (weekend news, halt reopens) are unhedged.
- Position selection ignores tax lot-level implications. Tax-lot management is out of scope.

2.8 Design Principles

Three non-negotiable filters govern every future proposal:

1. **Deterministic decides.** Every number reaching the broker comes from code in `src/`. No LLM output influences a size or weight.
2. **No magical thinking on knobs.** Every parameter traces to research or a repo-recorded backtest. Knobs whose provenance is intuition go on the research backlog.
3. **Slowly varying only.** Whipsaw is evidence of overfitting, not alpha. Long lookbacks are preferred over short. Drift thresholds err high. Regime signals (when added) require hysteresis and minimum dwell time. Factor timing is rejected on principle.

3 Parameters and Weights

3.1 The Factor Weights

The weight vector is momentum = 35%, quality = 30%, low-volatility = 20%, value = 15%. This is an *informed prior*, deliberately regularised away from any in-sample optimum.

3.1.1 Why not the in-sample winner?

The July 2026 diagnostic (grid search over the three-factor simplex, momentum/quality/low-vol, on the cached 2021–2026 history) produced in-sample winners uniformly momentum-heavy: 80–90% momentum, 0–30% quality, 10–20% low-vol. Sharpe up to 1.30 on the 3-year window versus ≈ 0.96 for the current policy.

This finding was deliberately not acted on. Three reasons:

1. **Single-regime evidence.** 2021–2026 was overwhelmingly momentum-friendly (post-COVID rally, mega-cap tech, AI narrative). Both the 5-year and 3-year windows overlap and sample the same regime. This is not an independent invariance test.
2. **Non-ergodicity.** Equity returns are multiplicative. For a single investor on a single path, the arithmetic mean of returns diverges from the time-average growth rate they actually experience (Peters 2019). A 50% drawdown requires 100% subsequent gain to recover, not 50%. In-sample-Sharpe optimisation understates this asymmetry. Factors

that reduce drawdown variance (quality, low-vol) buy protection against the multiplicative losses a single-path investor most needs to avoid. These are insurance premia, not underperforming factors.

3. **Confirmation bias.** The operator is explicitly on record as most concerned about an AI/tech valuation regime unwind. The mechanical winner (90% momentum) is exactly the exposure that would be worst affected in that scenario. Adopting the in-sample winner would be confirming a bias against the operator’s stated risk view.

3.1.2 The Constraint Framework

The 35/30/20/15 vector satisfies three constraints:

1. **Ordering: momentum > quality > low-vol > value.** Reflects the ordering of the empirical premia in the AQR literature and the compounding effect that the value implementation is currently a low-fee proxy rather than a true value factor.
2. **No factor below 10%.** A factor below 10% receives such a small weight in the weighted geometric mean that it fails to discriminate between candidates. The multi-factor-consistency requirement collapses to a three-factor screen.
3. **No factor above 40%.** A factor above 40% dominates the integrated score to the point that single-factor selection would produce nearly the same portfolio. This defeats the multi-factor rationale and re-introduces single-factor tail risk.

Within those constraints the exact values are round numbers chosen for auditability. They can be reproduced in one sentence to a future maintainer: “momentum gets the largest weight but capped, quality gets the second-largest, low-vol gets a solid third to hedge whipsaw, value gets the smallest until we fix the proxy.”

3.1.3 When To Revise

Revise the weights when at least one of the following is available:

- Backfilled history to pre-2020 that permits a non-overlapping regime comparison (2011–2015 vs 2021–2026).
- A fund-level P/E or P/B signal added to the value composite (yield + expense ratio blend committed 2026-07-10 partially satisfies this; full P/E/P/B would fully satisfy).
- A deliberate operator-declared factor-level sentiment override, committed to `configs/etf_smart_beta.toml` with a dated research note.

3.2 Portfolio Construction Parameters

Parameter	Value	Justification
num_positions	30	Target holding count. Prior operating configurations without a fixed target produced degenerate 1–2 position portfolios when factor scores were bimodal; the fixed target forces diversification. Raised from 20 to 30 on 2026-07-10 to spread capital more thinly across the top-ranked cohort and reduce single-name idiosyncratic risk in a sideways market.
min_weight	2%	Position floor. At 30 positions the full basket uses 60% of NAV at floor, leaving 40% for over-weight tilts on high-conviction positions.
max_weight	15%	Position cap. Prevents single-position concentration under the RankBased scheme; the top-ranked ETF cannot receive more than 15% of NAV.
tax_status	taxable	Signals real capital-gains consequences to any future tax-lot / wash-sale logic.

3.3 Factor Calculation Windows

Parameter	Value	Justification
Momentum lookback	252 days	Jegadeesh & Titman (1993). 12-month formation window.
Momentum skip	21 days	Novy-Marx (2012). Avoid short-term reversal.
Quality lookback	252 days	AQR convention.
Value lookback	252 days	Unused while value is expense-ratio proxy.
Volatility lookback	60 days	~3 months of daily returns. Standard practice.

3.4 Rebalancing Parameters

Parameter	Value	Justification
Scheduled frequency	bimonthly (~6/yr)	Balances signal freshness against turnover.
Drift threshold	5%	Round number, defensible per Donohue–Yip (2003). Widening (7.5%, 10%) considered before tightening.

3.5 Risk Management Parameters

Parameter	Value	Justification
Entry stop-loss	12%	Slightly wider than trailing stop to accommodate initial position volatility.
Trailing stop	10%	Sized to sit above ordinary noise ($\approx 2-3\sigma$ daily) so it triggers only on genuine trend breaks.
Cash reserve	\$30,000	Lowered from \$70k to \$30k on 2026-07-10. Roughly 8% of current NAV; freed \$40k for re-deployment into the 30-position basket. Static, not regime-responsive — automatic regime overlay tested and not enabled (§4).

3.6 Optimiser Priors (MVO variant only)

The live default optimiser is RankBased. The values below apply only when `OPTIMIZER_TYPE = "mvo"`.

Parameter	Value	Justification
Risk aversion λ	1.5	Standard MVO. Meucci retail range 2–4; 1.5 is risk-tolerant end.
Robustness penalty γ	0.7	Axioma-style, aggressive. Chosen default.
Turnover penalty	0.2	Medium. Chosen default.

4 Testing and Validation

The strategy has been subjected to a systematic multi-regime backtest with pre-registered acceptance criteria. This section replaces the prior “Current Flaws / adversarial review” framing: the identified concerns have either been addressed, tested and dismissed, or explicitly accepted after considered judgement. The purpose of this section is to establish, with numbers, that the operating policy is defensible.

4.1 Test methodology

Full write-up: `docs/research/2026-07_multi_regime_results.md` and companion CSV / plots. Headline discipline:

- **Universe:** 599-ticker curated smart-beta universe (`comprehensive_etf_list.py` categories minus commodities / currency / volatility products; leveraged and inverse ETFs excluded via `filter_universe` in the live pipeline and diagnostic alike). No filter-and-hope on niche exotica.
- **Data window:** 2011-01-28 to 2026-04-30, 3904 trading days (~ 15.5 years). Covers EU crisis 2011, taper tantrum 2013, China 2015–16, 2018 Q4, COVID 2020, 2022 rate cycle. Deep enough for a multi-regime verdict; not deep enough to claim ergodicity.
- **Windows:** results reported for full period, trailing 5-year, trailing 3-year. Robustness of a config is judged by min-metric across the three windows, not the peak in any one.

- **Pre-registered criteria:** weight change proposed only if a nearby config beats the prior by ≥ 0.10 Sharpe on the min-across-windows metric. Regime overlay enabled live only if Sortino improves *and* CAGR loss $\leq 2\%$. Clustering enabled only if MaxDD improves $\geq 2\%$ and CAGR loss $\leq 1.5\%$ across ≥ 2 windows. Written before results known; not retro-fitted.
- **Transaction cost model:** 5 bps applied to turnover per rebalance. A retail-plausible estimate; live cost is expected to be modestly higher.

4.2 Baseline performance

The current 35/30/20/15 policy, on the curated universe, over 15.5 years:

Window	CAGR	Sharpe	Sortino	Max DD	Days
Full (2011–2026)	13.04%	0.51	0.62	−36.9%	3 904
Trailing 5-year	12.44%	0.52	0.69	−19.1%	1 254
Trailing 3-year	19.31%	0.97	1.30	−13.4%	751

Context vs SPY same period: SPY CAGR $\sim 12.15\%$, Sharpe ~ 0.52 , MaxDD $\sim -34.1\%$. The strategy adds $\sim 0.9\%$ CAGR at slightly higher drawdown. A modest, defensible premium — not a home run, honestly reported.

Time-average log-CAGR: $+12.28\%$. Under non-ergodic compounding (see §3) this is the metric that actually accumulates over an operator’s investing horizon.

4.3 Factor weights: empirical validation of the prior

The 35/30/20/15 vector was chosen as an informed prior from AQR canonical evidence and the ergodicity argument. The empirical question: does a nearby config on the (momentum, quality, low-vol, value) simplex beat it enough to justify a change?

Config	min Sharpe	mean CAGR	Δ Sharpe vs prior	Note
2b_momentum_70	0.535	17.03%	+0.024	Empirical winner
2c_m40_q35_v15	0.518	15.49%	+0.007	Nearby-grid best
2a_volatility-0.10	0.512	15.02%	+0.001	Perturbation
...	...	...	...	(11 configs tied on plateau)
prior 35/30/20/15	0.511	14.93%	—	Reference
2b_quality_70	0.500	14.28%	−0.011	Loses
2b_volatility_70	0.241	9.17%	−0.270	Fails badly

Result: the prior sits inside a broad plateau of 11 configs sharing min-Sharpe within 0.001. The empirical winner exceeds it by +0.024 — well inside the pre-registered +0.10 threshold. The 70% momentum concentration wins narrowly (hinting momentum was the strongest single factor 2011–2026), but the multi-factor prior is preferred on drawdown-adjusted terms (Sortino) and on the design-principle grounds (§3: insurance premium, non-ergodicity).

Verdict: retain 35 / 30 / 20 / 15. The prior is not the empirical peak but is inside the plateau. No change proposed to `configs/etf_smart_beta.toml`.

4.4 Regime overlay: tested, not auto-enabled

An SPY-trend + VIX-level regime overlay (200-day SMA, 10-day hysteresis, 30-day dwell, 0.60 risk-off multiplier) was implemented (`src/portfolio/regime.py`), unit-tested for look-ahead safety, and backtested across five parameter combinations.

Config	CAGR	Δ CAGR	Sortino	Max DD	Δ MDD
no_regime (baseline)	13.04%	—	0.621	−36.9%	—
vix25_0.80	12.20%	−0.85%	0.624	−32.1%	+4.8%
vix25_0.60	11.26%	−1.78%	0.610	−27.1%	+9.8%
vix25_0.40	10.25%	−2.79%	0.567	−23.3%	+13.6%

The one config satisfying CAGR loss $\leq 2\%$ gains 0.003 Sortino ($\approx$ zero). On the trailing 5-year window — which includes the 2022 rate cycle, the regime the overlay should be most useful for — every regime config *underperforms* the no-regime baseline on both CAGR and Sharpe. The overlay’s protection was concentrated in early-window crises (2011, 2015–16); in modern data it costs performance for no drawdown protection.

Verdict: do not auto-enable. The module remains available as an operator-controlled defensive toggle to be applied by hand during acknowledged stress (e.g., a dated commit to `configs/etf_smart_beta.toml` setting an override flag). No automatic behaviour.

4.5 Correlation clustering: tested, rejected

A greedy pairwise-correlation cap on top-N selection (126-day rolling correlation window, quarterly refresh) was implemented (`src/portfolio/clustering.py`), unit-tested, and backtested across $\tau \in \{0.60, 0.70, 0.80, 0.90, 1.00\}$.

τ	CAGR	Sharpe	Sortino	Max DD	Verdict
1.00 (baseline)	13.04%	0.511	0.621	−36.9%	Reference
0.90	11.35%	0.464	0.552	−39.4%	Both worse
0.80	10.47%	0.432	0.511	−41.8%	Both worse
0.70	10.41%	0.426	0.508	−46.9%	Both worse
0.60	9.52%	0.373	0.443	−51.7%	Fails badly

Every clustering config makes both CAGR AND drawdown worse. The design assumption — that decorrelation buys drawdown protection — is not supported on this data. Mechanism: the top-scoring cohort is genuinely correlated (dividend-and-quality tilted defensives, or momentum-heavy growth in bull phases); forcing decorrelation pushes selection into lower-scored uncorrelated names that underperform.

Verdict: reject. The clustering module remains in the codebase for future re-examination (perhaps under different factor definitions or a stronger value tilt) but is not enabled and not proposed for enablement.

4.6 Sober response to prior adversarial concerns

A short, item-by-item accounting of what has been done since the earlier “Current Flaws” review:

- **Value factor was a fee proxy (T1.1).** Closed 2026-07-10. Real per-fund dividend yield + expense ratio blend from FMP Premium. Rankings verified (SCHD, HDV, VYM at the top; VUG, VGT, ARKK at the bottom). Fund-level P/E and P/B remain absent because no FMP tier publishes them for ETFs; the current implementation delivers the yield-based subset of the academic value composite. Not a fake factor any more.
- **Universe was uncured (implicit).** Now explicit. 599 hand-picked tickers, leverage and inverse enumerated, commodity / currency / volatility categories excluded. The diagnostic and live pipeline use the same screen. Prior backtests showing -68% and -52% drawdowns were artefacts of leveraged ETFs slipping through incomplete filters; the true baseline is -36.9% , in line with SPY.
- **Correlation blindness (T2.2).** Tested empirically. On multi-regime data, correlation clustering does not improve drawdown — it worsens it. The concern was reasonable; the empirical answer is that the current top-N-by-score behaviour is not the failure mode we feared.
- **Regime blindness (T2.1).** Tested empirically. Regime overlay’s protection was real in 2011–2015 crises but has disappeared in recent data. Available as a manual toggle; not automatic.
- **Survivorship bias (T1.2).** Accepted as small. The strategy’s momentum/quality screens filter delisting candidates long before they liquidate; residual bias is not worth the engineering cost to correct while the strategy proves out in live use.
- **Look-ahead risk (T1.3).** Accepted with spot-checks. Both the regime and clustering modules carry explicit look-ahead unit tests. Factor calculations rely on convention (rolling windows on lagged prices); the risk of a hidden $t+1$ dependency is judged low. No formal pinning test.

4.7 Open enhancements — not defects

Three items remain worth investigating, framed as improvements rather than flaws:

- **CAPE-conditional value tilt.** A regime-conditional weight adjustment based on external market valuation (e.g., trailing S&P 500 CAPE $\downarrow$ 30 $\Rightarrow$ tilt value up by 5%). Distinct from factor timing (see §3): the signal is market state, not factor performance. Requires CAPE data source verification and its own design note before implementation.
- **Fund-level P/E / P/B in value composite.** Would complete the academic value composite. Blocked on data source: FMP publishes neither at any tier; Morningstar Direct or FactSet would deliver it at institutional cost. Revisit after 3–6 months of live evidence.
- **Score magnitude as expected return.** Currently only rank order is used; exponential rank weighting expresses conviction indirectly. A small A/B against magnitude weighting (softmax on integrated scores) would settle whether the extra information helps. Runtime ~ 30 seconds; low priority but easy to run.

Factor lookback windows (252 / 21 / 60), rebalance cadence (bimonthly), and risk parameters (10% trailing stop, 12% entry stop, 5% drift threshold, 2–15% weight bounds) are inherited from convention and the pre-2026 backtest and are considered stable operating choices. Sensitivity work on any of these is a discretionary future investigation, not blocking.

5 Change Log

Chronological, most recent first. Each entry cites the closing commit SHA(s). Every commit touching `configs/etf_smart.beta.toml`, `src/factors/`, `src/portfolio/`, or the execution scripts must append an entry.

2026-07-10 — Multi-regime validation suite complete; Executive Summary rewritten with KPI headline

Full 15.5-year multi-regime backtest on the 599-ticker curated universe, five parameter dimensions, pre-registered acceptance criteria. Documented in `docs/research/2026-07_multi_regime_results.md`. Baseline (35/30/20/15) delivers CAGR +13.04%, Sharpe +0.51, Sortino +0.62, MaxDD -36.9%; alpha vs SPY +0.89%/yr. Weight prior validated (empirical winner +0.024 Sharpe, threshold was +0.10); regime overlay rejected for auto-enable (Sortino gain +0.003 at CAGR cost -0.85%; underperforms baseline on trailing 5-year window); correlation clustering rejected (all configs made both CAGR and MaxDD worse). No config changes proposed. Universe screen (`filter_universe(use_curated=True)`) enforced identically in live pipeline and diagnostic. Executive Summary and Testing sections rewritten to lead with the headline KPIs. Commits 61ba98a (curated screen), 29b2efd (results).

2026-07-10 — T1.1 partial close: real value factor via FMP Premium

Value factor rewired end-to-end. `src/factors/value_factor.py` now blends real per-fund dividend yield (60% weight) with expense ratio (40% weight). Fundamentals cache at `~/trade_data/ETFTrader/pr` refreshed weekly by `scripts/refresh_fundamentals.py` over the union of the defined universe and the price cache (~5000 tickers). Data source is Financial Modeling Prep Premium tier (~\$44/mo; verified capabilities on 2026-07-10 — see `rule_verify_before_recommending` memo). Sanity check: SCHD scores 1.04, VYM 0.56, VTI -0.05, VUG -0.39, ARKK -1.81 — the expected value ordering. Residual: fund-level P/E and P/B remain unavailable at current FMP tier, documented in T1.1. Commit: 2d4c671 (endpoint verification), plus follow-up wiring.

2026-07-10 — Position count 20→30, cash reserve \$70k→\$30k

Operator directive following observation that the current 20-name basket has been drifting sideways with the market: raise `num_positions` to 30 and lower `cash_reserve` to \$30,000 in `configs/etf_smart_beta.toml`. Rationale: spread capital more thinly across the ranked top-cohort to reduce single-name idiosyncratic drag, and free approximately \$40k of dormant cash for redeployment. Passes the slowly-varying filter (parameter change, not signal change) and preserves the design invariants ($30 \times 2\%$ floor = 60% < 100%; ranking and scoring logic untouched). Related open item: T2.2 correlation clustering — with a wider basket the marginal decorrelation benefit from clustering is expected to fall, and the experiment should quantify this.

2026-07-10 — Full rewrite

Document rewritten from the ground up. Historical version-comparison content (v3 → v4 migration story, v4.2 backtest tables, Feb-2026 holdings snapshot) removed. New structure: Executive Summary → Strategy Mechanics → Parameters and Weights → Current Flaws → Change Log. Factor definitions and integration rationale promoted into the Mechanics section; the constraint framework (10%–40% weight bounds, factor ordering) promoted into the Parameters section. Length approximately halved.

2026-07-09 — Adversarial review + weight justification (superseded by rewrite)

Adversarial review commissioned and recorded. Ergodicity-economics justification for retaining the 35/30/20/15 weight vector recorded in the Factor Framework section. Design Principles (deterministic decides, no magical thinking, slowly varying) recorded as non-negotiable filters. Level-1 factor-weight diagnostic produced at `docs/research/2026-07_factor_weight_diagnostic.md`. Commits d9826dc, 2198273, 6a14212.

2026-02-15 — Prior v4.2 design (historical, retained for context)

Prior design documented 30 positions, 3–8% weight bounds, quarterly rebalance, Robust MVO default. Backtested 12.1% CAGR / 0.64 Sharpe over 2021–2026. Live configuration diverged from this design in three places (position count, weight bounds, default optimiser) and the 2026-07 rewrite reconciled the document to what is actually running.